11-10 第1篇

阅读理解（说明文） Why Do We Get Goose Bumps?

① On the first cold morning of November, my little sister stepped outside, pushed up her sleeve, and laughed. "Look at my arm!" she said. The hair on her forearm was standing up, and on the surface of her skin there were **tiny, round hills**, each one at the bottom of a hair. Chinese people call them **鸡皮疙瘩** ("chicken-skin bumps"); English-speaking people call them **goose bumps** (after the rough skin of a cleaned goose). Almost everyone has had them, yet almost no one can explain them. Where do those tiny hills come from, and what are they trying to do?

② To understand goose bumps, we need to look at the skin under a microscope. Every tiny hair on our body grows from a small pit in the skin, called a **hair follicle**. Attached to each follicle is a **very small muscle** with a long Latin name — *arrector pili* — which simply means "the muscle that lifts the hair". When the brain sends a quick signal, this tiny muscle **pulls the hair upright**. Around the base of the hair, the skin bunches up into a tiny round bump. Multiply this by millions of hairs, and the whole arm seems to rise in small hills in less than a second.

③ Why would the brain send that signal? The answer lies **deep in the past**. Thousands of years ago, our far, far ancestors — and even earlier, animals that looked nothing like us — were covered with thick fur. When a wolf feels cold, it lifts its fur so that **a thicker layer of still air** is trapped close to the skin. Still air is a very good **insulator**, which means it blocks the movement of heat. The wolf's body, warmed by its blood, is now wrapped in a fluffy warm blanket of its own making. When a cat meets an enemy, it fluffs up the same way so that it **looks bigger and scarier**. Both actions are run by the same tiny muscle at each hair.

④ Human beings have kept the muscle, but mostly **lost the fur**. When we step out on a cold morning, the brain still sends the old signal; the tiny muscles still pull; the hairs still rise. But there are too few and too short to keep any warmth in, and far too few to make us look bigger. The result we see is only the side-effect: **tiny bumps of skin**, and hair you have to look closely to notice. Goose bumps are, in other words, a **very old reflex** left behind on a modern skin — something like a button on an old coat that you no longer use, but cannot bear to cut off.

⑤ Goose bumps do not only come from cold. They also appear when we feel **strong emotions**: fear at a dark corridor, shock when a car horn suddenly sounds, joy when a familiar song returns on the radio, or even a quiet moment when a grandmother tells a long story. In these moments, the **nervous system** releases a small burst of a chemical

called **adrenaline**. Adrenaline squeezes many small parts of the body — the heart beats faster, the pupils widen, and — yes — those tiny hair-muscles tighten. This is why some people say "**music gave me goose bumps**". Strictly speaking, it was the adrenaline, responding to the music.

⑥ Are goose bumps useful to us at all? Mostly they are a small, harmless reminder that our body is **old**, older than our jeans and our phones. But even today doctors sometimes read them as a gentle signal. A sudden wave of goose bumps without cold or strong feeling may mean a **fever is starting**. Unusually frequent goose bumps together with shaking hands may be a small sign of fear that the person has not yet said out loud. The body, in its quiet way, sometimes speaks a sentence before the mouth does.

⑦ My sister, back in the warm kitchen, rubbed her arm and watched the tiny hills go down. "So it's like my skin is saying **'I remember I was once a wolf'**?" she asked. I laughed. It is not a bad image. On every cold morning, for a few seconds, a very old story runs across our arms — a fur, a cold wind, a wild night. Then the body quiets down, the tiny hills go flat, and we pick up our schoolbag and walk on. Without knowing it, **we carry the winter of our ancestors inside us, and it still knows how to speak**.

注释： goose bumps 鸡皮疙瘩；follicle 毛囊；insulator 绝缘体；adrenaline 肾上腺素；reflex 反射

根据短文内容，选择最佳答案。

1. What do we call the small muscle that makes the hair rise?

A. A big skin muscle.
B. Every hair grows from a pit called a *hair follicle*; attached to each follicle is a tiny muscle called *arrector pili*, which means "the muscle that lifts the hair".
C. A stomach muscle.
D. A heart muscle.

2. Why is goose bumps a USEFUL reaction for a wolf but NOT for a human in the cold?

A. Because wolves are faster than humans.
B. Because a wolf has thick fur: when the hair rises, a layer of still air is trapped as a warm blanket; humans have too few and too short hairs to trap enough warm air.
C. Because humans do not feel cold.
D. Because wolves do not have skin.

3. How does strong emotion cause goose bumps, according to Paragraph ⑤?

A. Emotion directly squeezes the skin.
B. The nervous system releases adrenaline, which tightens many small parts of the body

— including the tiny hair-muscles — so we can get goose bumps from fear, shock, joy, or quiet moments.
C. Emotion makes the hair fall out.
D. Emotion changes the colour of the hair.

4. Which of the following might doctors sometimes read as a GENTLE signal from goose bumps?

A. A student's height is increasing.
B. A fever is starting, or a person is afraid but has not yet said so.
C. The room is too large.
D. The student is hungry.

5. Which sentence best describes the author's feeling towards goose bumps in Paragraph ⑦?

A. Goose bumps are a useless, ugly mistake.
B. Goose bumps are a small, quiet doorway through which a very old story of our animal past — a fur, a cold wind, a wild night — runs across our arms; we carry the winter of our ancestors inside us.
C. Goose bumps are only a skin illness.
D. Goose bumps are always dangerous.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段 "hair follicle ... arrector pili ... the muscle that lifts the hair"
【解析】解剖结构直引。A、C、D 均错。故选 B。

2.
【答案】B
【题型】推理判断题
【定位】第③、④段对比
【解析】保暖原理差异。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段 "adrenaline ... tightens ... hair-muscles"
【解析】肾上腺素机制。A、C、D 均错。故选 B。

4.
【答案】B

【题型】细节理解题
【定位】第⑥段 "fever is starting ... fear not yet said out loud"
【解析】医学线索直引。A、C、D 均错。故选 B。

5.
【答案】B
【题型】推理判断题
【定位】第⑦段末 "winter of our ancestors inside us"
【解析】作者态度温柔而富感情。A、C、D 均偏离。故选 B。